

EHR Integration Product Requirements

Problem: Double entry and managing two sources of data: the EHR and Viewpoint Comanagement.

This integration will automatically pull data from the **Nextech** and **NextGen** EHR systems and create/update it in Viewpoint Comanagement. This is to reduce the double entry friction users will face and rely on the EHR as the single source of truth for overlapping data.

The data that is pulled must be reviewed and then potentially added to our database. We need to be able to fetch clinic information (info about the building/practice the providers work at), provider information (contact info, name, etc for optometrists), patient information (name, DOB, and anything else we can get) electronically gathered from the ophthalmologist practice/user of the EHR.

Obstacles: Multiple versions of the Nextech software plus NextGen has a different API setup. Both however, have decent documentation. The versions of the Nextech software: NexCloud/Select, Practice +, IntelChartPRO, or SRS Pro.

NexCloud/Select and **Practice +** have the same API endpoints and associated parameters.

Since there are no defining fields on the patients that clearly distinguish it as a comanaged patient, with the EHR integration we need to be able to fetch ALL patients. Then from there, once a day we can pull ALL patients that have been created since the last fetch. E.g. Initial fetch for when the user onboards to the platform. Of those patients some need to be defined as comanagement patients. Those that are not are dropped, those that are, are saved into the db.

Add EHR selector

1. MUI Select component that is a dropdown with the options to select being the current EHR integrations we have.
 - a. Since we don't have any existing integrations then it will default to the NexCloud/Select EHR.
 - b. Save this value on the `organizations` table in the `ehrVersion` field.

Referrals - Fetch all patients from Nextech EHR

HTTP Request

`GET` - see Nextech versions and NextGen

1. Compare the fetched result to existing `patients` in the database and see which fields have been changed.
 - a. Update the corresponding fields in my database that have been changed in the EHR.
 - b. This should be performed as a cron job that runs once every 24 hours.
2. If the result of the query indicates new patients (aka they don't already exist in the db) then display those results of the new patients to the user in a table on the Action Items page [MUI DataGridPro](#)
 - a. each row will show some brief patient data and have actions associated with it. This is to determine whether this patient is a comanaged patient.
 - i. Display patient's `name`, `birthDate`, `patient-note`, `generalPractitioner`, `referring-physician.HumanName`, `patient-status`, `patient-type` and referring clinic (if not available from EHR, then needs to be determined based on the clinics associated with the provider who is the referring provider)
 - ii. NOTE: the patient *must* have a referring provider and referring clinic set in order to be successfully saved in the db.
 - iii. The table should be in-line editable (double click the cell) to allow a user to set the referring provider and set the referring clinic. Those columns should have a dropdown with all available selections listed to choose from
 - iv. Have the referring provider column be editable/selectable first. Then the referring clinic column will change selection options and only show the clinics that the provider is associated with. If the user clicks on the clinic column to edit it and there is no referring provider set it should be an empty dropdown and have some small text indicating to "Select referring provider first".

v. Rows that don't have a referring provider AND a referring clinic set should be easily discernible at a glance by having the background color of the row to a soft red. Once BOTH referring provider and referring clinic are set, then the background color of the row changes to a normal, neutral color.

b. Hopefully, based on the above table data (can add more fields later), that will be enough to make a determination of whether the patient is a comanaged patient or not. By default the patients will be assumed to *not* be comanaged. Therefore the user must go through each row and determine if it *is* comanaged.

i. Each row should have a checkbox in the farthest left column that when clicked will slightly change the background color of the row to indicate a selection (in addition to the checkbox showing checked)

ii. There will be a button in the toolbar area of the data table labeled "Add Referrals", that a user can click to insert all of their selected patients into the database.

1. If there is a selected row that does not have a referring provider and referring clinic set then stop the attempt to save and indicate why via a snackbar notification. "Unable to save. Please select a referring provider and referring clinic for the referral". It is a frontend validation.

2. If all rows to be saved have proper referrers set then there should be a confirmation dialog box that pops up on click that verifies the user wants to proceed to add {count} of patients to the database.

3. Once the affirmative is selected from the confirmation dialog box, select patient data (that is mapped below) will be saved into our db. Use `ADD_PATIENTS` query for the GQL mutation.

4. Upon successful save of those patients the data table will become empty. Discard both comanaged and not comanaged patients in that table.

5. Those successfully saved referrals will now be displayed in the Referral Table on the Referrals page.

c. There should be a track of the count of the number of rows still enabled (which indicate number of comanaged patients) and that should be displayed at the top above the table. Either in the toolbar area or above the table itself.

d. Once the selection of rows is finished there needs to be an action taken where the user can insert these new patients into our database.

i. The button should be in the toolbar of the data table.

e. Each time the patients are fetched, save the date of the most recent referrals fetch on the `organizations` table.

`dateLastReferralsFetch`

3. When to fetch the referrals

a. Cron job

i. Run this once per day well after the workday, like just before midnight

ii. Upon successful fetch, save the potential new referrals in the db on the `potentialReferrals` table. I need to see which fields exactly we'll need to save once I understand the Nextech API better

b. Button click

i. I need a manual way of fetching referrals if that is so desired. Put this button on the Action Items page. Put text next to the button indicating when the last fetch was. `Last fetch: Yesterday 09/12/24`

For initial onboarding and implementation to the software we need to fetch ALL existing patients in the EHR

1. Display the results of the fetch in a data table as described in the above `GET` query.

2. Determine which patients are, or will be, comanaged. Do this in the same way as described above.

3. Allow row editing as described above.

Clinics - Fetch all clinics

`GET` see Nextech and NextGen associated documentation

1. Results of fetched the data should be compared to the existing data in the `clinics` table.
 - a. There isn't a way to fetch the `Location` using an `lastUpdated` parameter so it seems the only way to recognize if a new Location was added to the EHR and needs to be added to our db is to fetch all the Locations and compare both the names and addresses to see if they are the same. Some clinics have the same name but different addresses.
 - b. Additionally, to check for any need to update, once the fetched clinic locates its existing twin in my db it needs to then compare every field to determine if an update needs to happen. I.e. Compare any changes to the EHR fields that need to be adjusted in my db.
 - c. This would also be a cron job that runs once per day
 - d. This needs to be run as part of onboarding/implementation as well. Not sure how to do this? Through a curl operation?
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Providers - Fetch all providers

GET see Nextech and NextGen associated documentation

1. Results of fetched data should be compared to existing data in the `providers` table.
 - a. Same issue as the Location-clinics. No way to fetch the practitioner data by `lastUpdated` parameter. Need to know if there are any new practitioners added to the EHR and then add them to my db and need to compare existing records to their EHR twin and see if any EHR fields have changed that also need to be updated in my db
 - b. This would also be a cron job that runs once per day
 - c. This needs to be run as part of onboarding/implementation as well. Not sure how to do this? Through a curl operation?
 2. Providers who have email addresses will be created as `users` with the role of `optometrist`
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Appointments - fetch all appointments

GET see associated Nextech documentation

1. Results of fetched data will include dates, description and participants in the appointment
 2. The participant field will need to be looped over to determine who the patient is. Then querying for that patient's id in my database, updating its `surgeonId` and `clinicId` to reflect the remaining values: the location and practitioner names.
 3. Most likely will have to query my db for the `provider` (in this case the surgeon) in order to connect the `surgeonId` and query the `clinics` table to obtain the id in order to connect it to the patient as the `clinicId`.
 4. Utilizing the description of the appointment, determine if this is an initial consult or surgical appointment. If it is an initial consult set the date of the appointment on the `dateInitialAppointmentScheduled` on the `patientContacts` table.
 - a. If this is a surgical date, check if there already is a date set on the `preOperations` table for the `firstEyeSurgeryDate` field. If not, set the date on this field. If there is a date already on the `firstEyeSurgeryDate` field and it's in the past then set the date on `secondEyeSurgeryDate`.
 - b. If there is confusion as to which surgery date (first or second) should be set then do not set one and save free text on the `contactNotes` field on the `patientContacts` table that says, "Verify in the EHR what surgery dates this patient has"
 5. Save appointment type
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Handling creation errors

There is a need to handle any errors of failed creations of referrals, clinics and providers as a result from the cron job.

I am open to thoughts here, but one option could be saving those records that failed to save in our db as an array on the field `potentialAdditionFailures` on their respective table. E.g. 4 records of referrals fail to save properly. Their fetched data is saved as an array of objects on the `potentialAdditionFailures` on the `patients` table or `clinics` table or `providers` table. (Note: this field does not exist yet on those tables)

Deliverables

- functions that fetch the required data
 - make it so these functions are triggered by the user from a button
 - Cron jobs that run once a day just before midnight (or well after the work day has concluded) to automatically fetch data and make updates
- logic that will review the fetched data and compare it to existing records in the database
- logic that then updates or creates records in the database based on the fetched data
- written step-by-step process for how to work with NexTech and obtain necessary requirements (practice identifier) in order to have a new user have an EHR integration.
- Cron jobs that run once a day just before midnight (or well after the work day has concluded) to automatically fetch data and make updates